CL469 - Assignment Two: LBM Simulation of Poiseuille Flow

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1 Introduction

This report details the simulation of 2D gravity-driven Poiseuille flow using the Lattice Boltzmann Method (LBM) as per Assignment Two for CL469. The simulation models water flow between parallel plates under constant gravity, employing the D2Q9 lattice model. Key objectives include verifying the steady-state velocity profile against the analytical solution, calculating wall forces using multiple methods (Momentum Exchange, Stress Integration, Finite Difference), investigating the effect of the relaxation parameter τ and the collision operator (BGK vs. TRT) on wall slip, and comparing simulated viscous dissipation with theoretical values. The governing Lattice Boltzmann equation with external forcing is solved numerically.

2 Simulation Setup

2.1 Lattice and Flow Parameters

The simulation domain consists of a 2D lattice with dimensions $N_x \times N_y$. Periodic boundary conditions are applied in the x-direction (length $L_x = N_x \Delta x$) and no-slip bounce-back boundary conditions are applied at the top ($y = N_y - 1$) and bottom ($y = 0$) walls (channel height $H = (N_y - 1)\Delta y$). For simplicity, we use $\Delta x = \Delta y = \Delta t = 1$. The reference case uses $N_y = H = 11$ (meaning $H_{width} = 10$ lattice units) and $N_x = L_x = H = 11$. The simulation parameters are chosen to match the physical Reynolds number ($Re_p = 0.1531$) for the reference case:

- Relaxation time (BGK): $\tau = 0.8$
- Kinematic viscosity: $\nu = c_s^2(\tau - 0.5\Delta t) = (1/3)(0.8 - 0.5) = 0.1$
- Body force (gravity): $g_x = 9.203 \times 10^{-6}$ (chosen to yield $Re = 0.1531$ for $H = 11$)
- Initial density: $\rho_0 = 1.0$
- Lattice speed of sound squared: $c_s^2 = 1/3$

Simulations are run until a steady state is reached, typically determined by monitoring the L2 norm of the relative velocity change between timesteps, or for a maximum number of steps (e.g., 15000). The viscous time scale $t_\nu = H_{width}^2/\nu = 10^2/0.1 = 1000$, suggesting simulations should run significantly longer than this.

2.2 Numerical Implementation

The C++ code implements the D2Q9 LBM algorithm. Initialization sets $\mathbf{u} = 0$ and $\rho = \rho_0$, with populations $f_i = f_i^{eq}(\rho_0, \mathbf{u} = 0)$. The main loop performs:

1. Calculation of macroscopic density ρ and velocity $\mathbf{u}$ (including force term $0.5\mathbf{F}\Delta t/\rho$) from current populations f_i.

2. Collision step: Calculates post-collision populations f_i^* using either the BGK operator (Eq. 26) or the TRT operator (Eq. 33). The Guo et al. forcing scheme (Eq. 27) is incorporated to account for gravity $\mathbf{F} = \rho\mathbf{g}$.

3. Streaming step: Propagates f_i^* to neighboring nodes (Eq. 28).

4. Boundary conditions: Applies periodic BC in x and bounce-back (half-way) at y=0 and y=Ny-1.

Wall forces are calculated using Momentum Exchange (Eq. 22), Stress Tensor Integration (Eq. 30, using stress tensor from Eq. 17/Textbook 5.17), and Finite Difference (Eq. 31). Viscous dissipation is calculated using Eq. 37. Parameter studies are automated using the ‘run$_{p}arameter_{s}tudy.sh‘script, andvisualizationsaregeneratedusingPythonscriptsinthe‘visualizations/‘dire$

3 Results and Discussion

3.1 Steady State Velocity Profile

The simulation was run with the reference parameters ($H = 11, \tau = 0.8$). The steady-state velocity profile $u_x(y)$ at the channel center ($x = N_x/2$) is compared with the analytical solution (Eq. 5) in Figure 1. The velocity is non-dimensionalized by the analytical maximum velocity $u_{max} = gH_{width}^2/(8\nu)$.

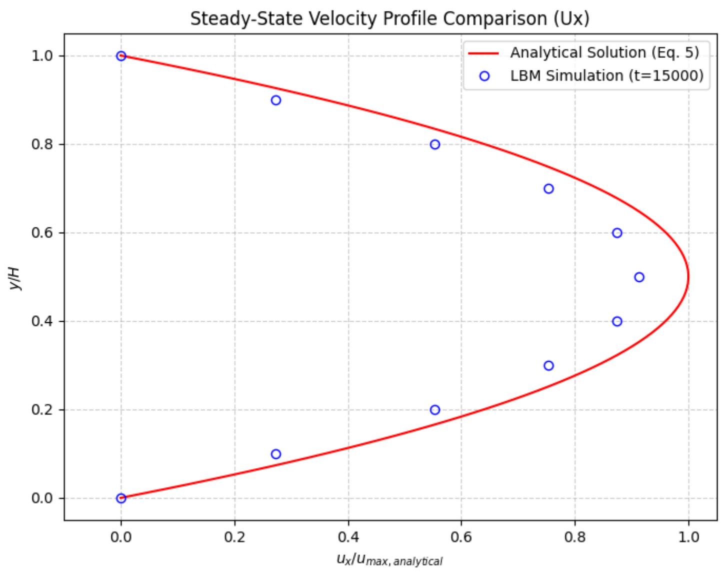

Figure 1: Comparison of simulated (dots) and analytical (line) steady-state velocity profiles, non-dimensionalized.

Discussion: (Comment on the agreement between simulation and theory. Report the maximum observed u_y and confirm it's negligible using Figure 2. Comment on the density profile shown in Figure 3 - is it uniform as expected for incompressible flow?)

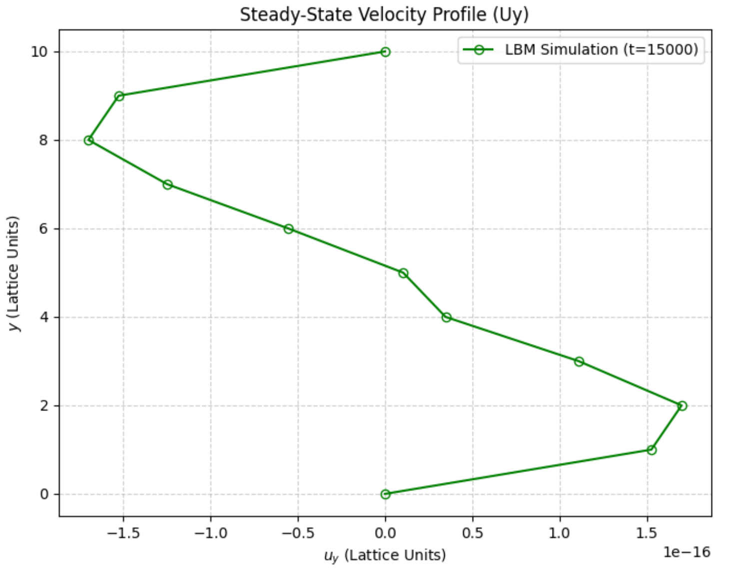

Figure 2: Simulated steady-state u_y velocity profile.

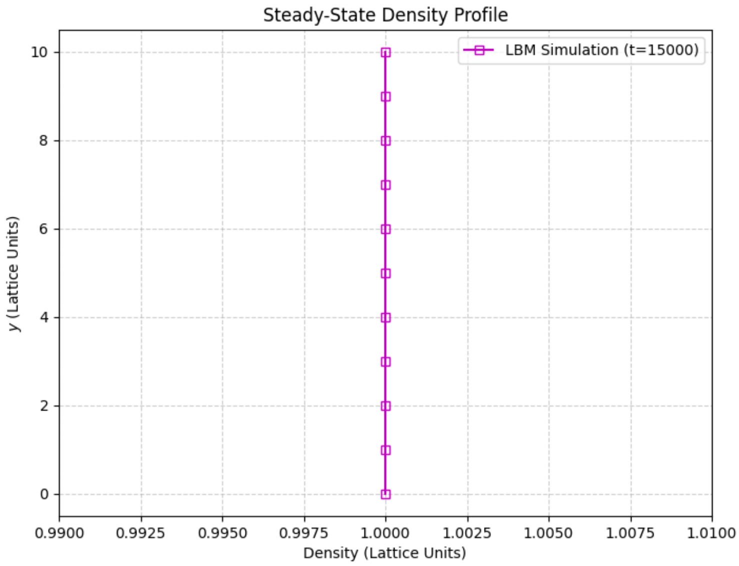

Figure 3: Simulated steady-state density profile.

3.2 Wall Forces: Momentum Exchange

The forces acting on the top and bottom walls were calculated using the Momentum Exchange method (Eq. 22) throughout the simulation. Figure 4 shows the non-dimensional x-component of the force ($F_x/W_{lattice}$, where $W_{lattice} = \rho_0 g_x N_x H_{width}$) versus time. The analytical non-dimensional force is 0.5 (Eq. 8).

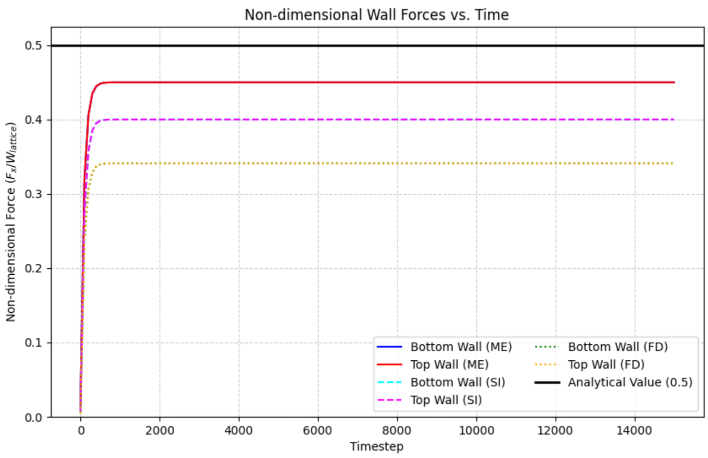

Figure 4: Non-dimensional x-force on top and bottom walls calculated using different methods vs. time for the reference case ($H = 11, \tau = 0.8$). The black line indicates the analytical value of 0.5.

Discussion: (Comment on the steady-state value achieved by ME compared to 0.5. Discuss the y-component of the force shown in Figure 5 - is it negligible as expected? Figure 6 compares the convergence of the different x-force calculation methods for the reference run. Raw force values are shown in ‘figures/forces$_v s_t ime.png$‘.)

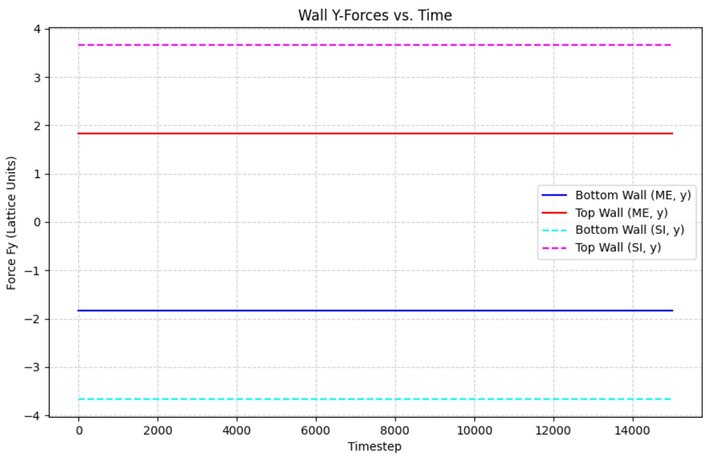

Figure 5: Y-component of forces on top and bottom walls calculated using different methods vs. time for the reference case ($H = 11, \tau = 0.8$).

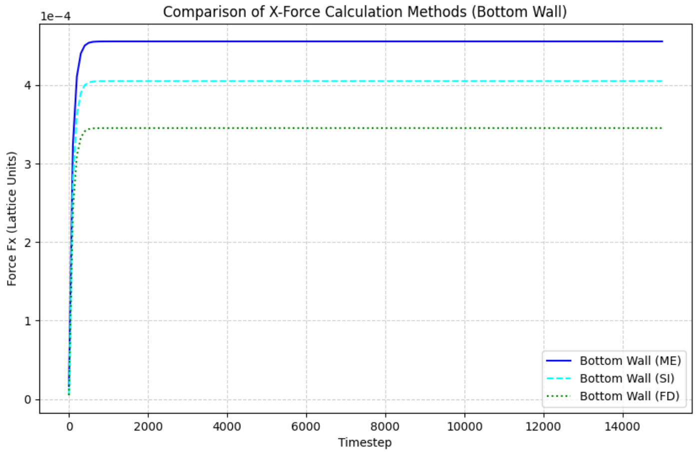

Figure 6: Comparison of X-force calculation methods (ME, SI, FD) on the bottom wall vs. time for the reference case ($H = 11, \tau = 0.8$).

3.3 Mach Number Independence

(This section requires manual runs and analysis, not automated by the provided script). The assignment asks to verify that the non-dimensional force is independent of the Mach number ($Ma = u_{max}/c_s$) for this incompressible flow. This involves running simulations with different u_{max} (achieved by varying g or grid resolution H) while keeping the Reynolds number constant, and showing that the non-dimensional force remains close to 0.5.

3.4 Viscous Stress Tensor

(This section requires adding specific output or analysis to the C++ code). The assignment asks to calculate the viscous stress tensor τ (Eq. 17/29) and verify that τ_{xx}, τ_{yy} are negligible and $\tau_{xy} = \tau_{yx}$.

3.5 Wall Forces: Comparison of Methods

To assess the accuracy of different force calculation methods, simulations were run for varying channel resolutions $H = 5, 10, 20, 40$ (keeping other reference parameters fixed). The relative percentage error of the non-dimensional x-force compared to the analytical value of 0.5 ($\epsilon = (F_{x,nd} - 0.5)/0.5 \times 100\%$) is plotted against H for the Momentum Exchange, Stress Integration, and Finite Difference methods in Figure 7.

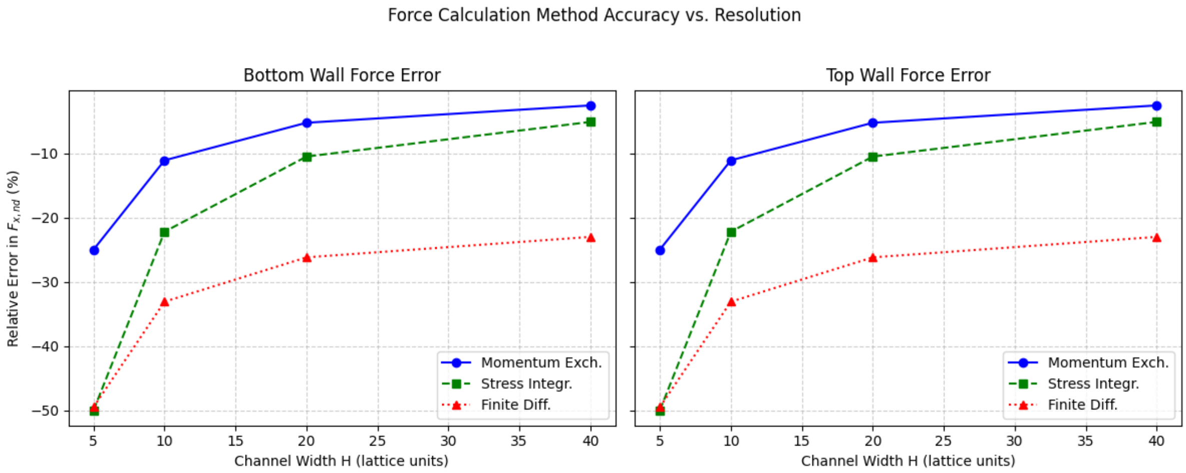

Figure 7: Relative percentage error in non-dimensional x-force vs. channel resolution (H) for different calculation methods (ME, SI, FD) on the bottom and top walls.

Discussion: (Analyze the convergence and relative accuracy of the three methods as resolution increases. Which method performs best?)

3.6 Effect of Relaxation Parameter and TRT

The standard BGK collision operator coupled with halfway bounce-back can lead to unphysical slip velocity at walls for large relaxation times τ. To demonstrate this, simulations were run with the reference $H = 11$ but using $\tau = 2.0$ and 5.0. The Two Relaxation Time (TRT) operator aims to mitigate this. Simulations were also run using TRT (with $\tau^- = 1.0$) for $\tau = \tau^+ = 2.0$ and 5.0. Figure 8 compares the near-wall velocity profiles for BGK and TRT at these high τ values.

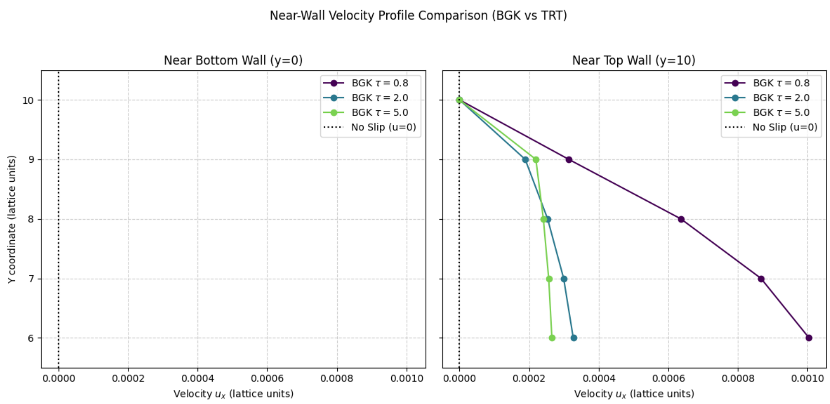

Figure 8: Comparison of near-wall velocity profiles for BGK and TRT collision operators at high relaxation times ($\tau = 2.0, 5.0$).

Discussion: (Analyze the plots in Figure 8. Does BGK show significant slip at the bottom (y=0) and top (y=H) walls for high tau? Does TRT reduce or eliminate the slip at both walls?)

3.7 Viscous Dissipation

The total viscous dissipation W_d was calculated from simulations using Eq. 37 for different channel resolutions ($H = 5, 10, 20, 40$). This is compared with the analytical value derived from integrating $\mu(du_x/dy)^2$ over the volume, which gives $W_{d,analytical} = (\rho g_x^2 L_x H_{width}^3)/(12\nu)$. Figure 9 shows the comparison.

Analytically, the work done by gravity per unit time is $W_g = \int_V \rho \mathbf{g} \cdot \mathbf{u} dV = \int_0^{L_x} \int_0^{H_{width}} \rho g_x u_x(y) dy dx$. Substituting $u_x(y) = \frac{g_x}{2\nu} y(H_{width} - y)$ and integrating yields $W_g = \rho g_x L_x \frac{g_x}{2\nu} [\frac{H_{width} y^2}{2} - \frac{y^3}{3}]_0^{H_{width}} = \frac{\rho g_x^2 L_x H_{width}^3}{12\nu}$, which exactly matches $W_{d,analytical}$.

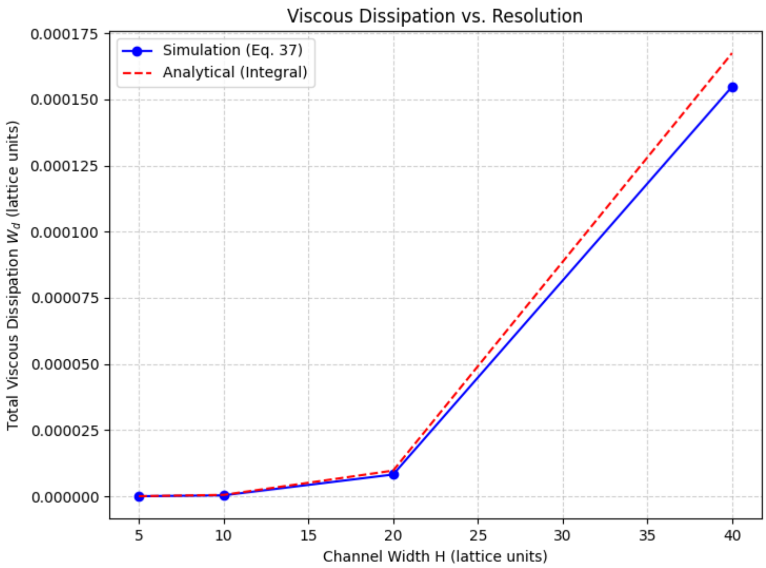

Figure 9: Comparison of simulated viscous dissipation (Eq. 37) with the analytical value vs. channel resolution H.

Discussion: (Does the simulated dissipation match the analytical value according to Figure 9? How does the accuracy (relative error, Figure 10) change with resolution?)

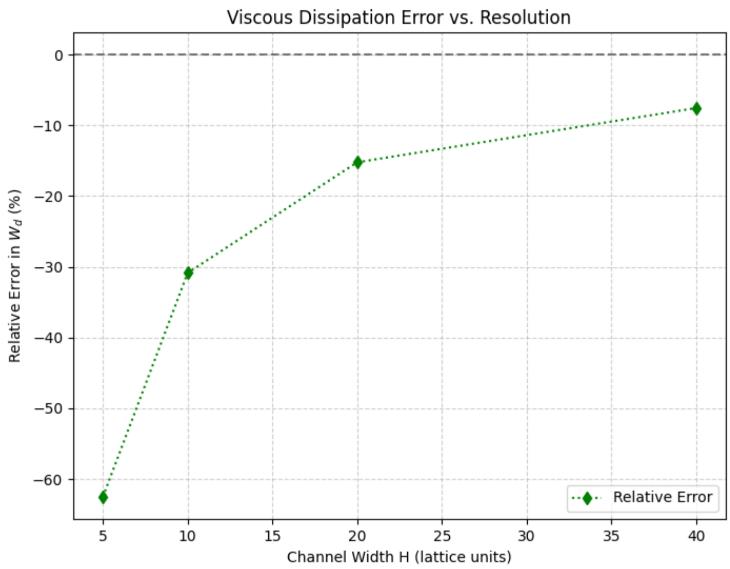

Figure 10: Relative percentage error in simulated viscous dissipation vs. channel resolution H.

4 Conclusion

Summarize the key findings. Was the velocity profile accurately reproduced? How did the different force calculation methods perform? Was TRT effective in reducing slip? Did the simulated dissipation match the theory and the work done by gravity?

References

List any external resources consulted (textbook, papers, online resources, AI tools like ChatGPT, Gemini Code Assist, etc.). Refer to the assignment PDF for specific references (Krüger et al., Guo et al., Latt et al.).

- Krüger, T., Kusumaatmaja, H., Kuzmin, A., Shardt, O., Silva, G., Viggen, E. M. (2017). *The Lattice Boltzmann Method: Principles and Practice.* Springer.
- Guo, Z., Zheng, C., Shi, B. (2002). Discrete lattice effects on the forcing term in the lattice Boltzmann method. *Physical Review E*, 65(4), 046308.
- Assignment Two PDF: CL 469 - Lattice Boltzmann method for fluid flow (Date: 31-03-25)
- (Add any other resources used, e.g., specific websites, AI tools)

A Source Code Snippets (Optional)

Key parts of the implementation can be included here using the 'listings' package.

Example: Collision step implementation

```
// ... inside collideBGK loop ...
 equilibrium(rho_node, ux_node, uy_node, feq_node.data());

 for (int i = 0; i < LBM::Q; ++i) {
     size_t pop_idx = idx(x, y, i);
     double Fi = 0.0;
     calculateForceTermBGK(i, rho_node, ux_node, uy_node, Fi);

     // Eq. 26 (BGK collision + Guo force term)
     f_new[pop_idx] = f[pop_idx] * (1.0 - params.omega) + params.omega *
         feq_node[i] + Fi;
 }
// ...
```

Listing 1: BGK Collision with Guo Forcing in Simulation::collideBGK

Example: Momentum Exchange Force Calculation

```
// ... inside calculateForcesMomentumExchange loop ...
 // Bottom wall (y=0): Consider fluid nodes at y=1
 for (int i = 0; i < LBM::Q; ++i) {
     int fluid_y = 1;
     if (LBM::cy[i] < 0) { // Directions from fluid (y=1) to wall (y=0) (i=4,
         7, 8)
         size_t pop_idx = idx(x, fluid_y, i); // f*_i at node (x, 1)
         double f_star_i = f_new[pop_idx]; // Use post-collision f_new

         // Based on momentum transferred across the link per step
         F_bottom_x += LBM::cx[i] * f_star_i;
         F_bottom_y += LBM::cy[i] * f_star_i;
     }
 }
// ...
```

Listing 2: Momentum Exchange Force Calculation in Simulation::calculateForcesMomentumExchange (Excerpt for Bottom Wall)